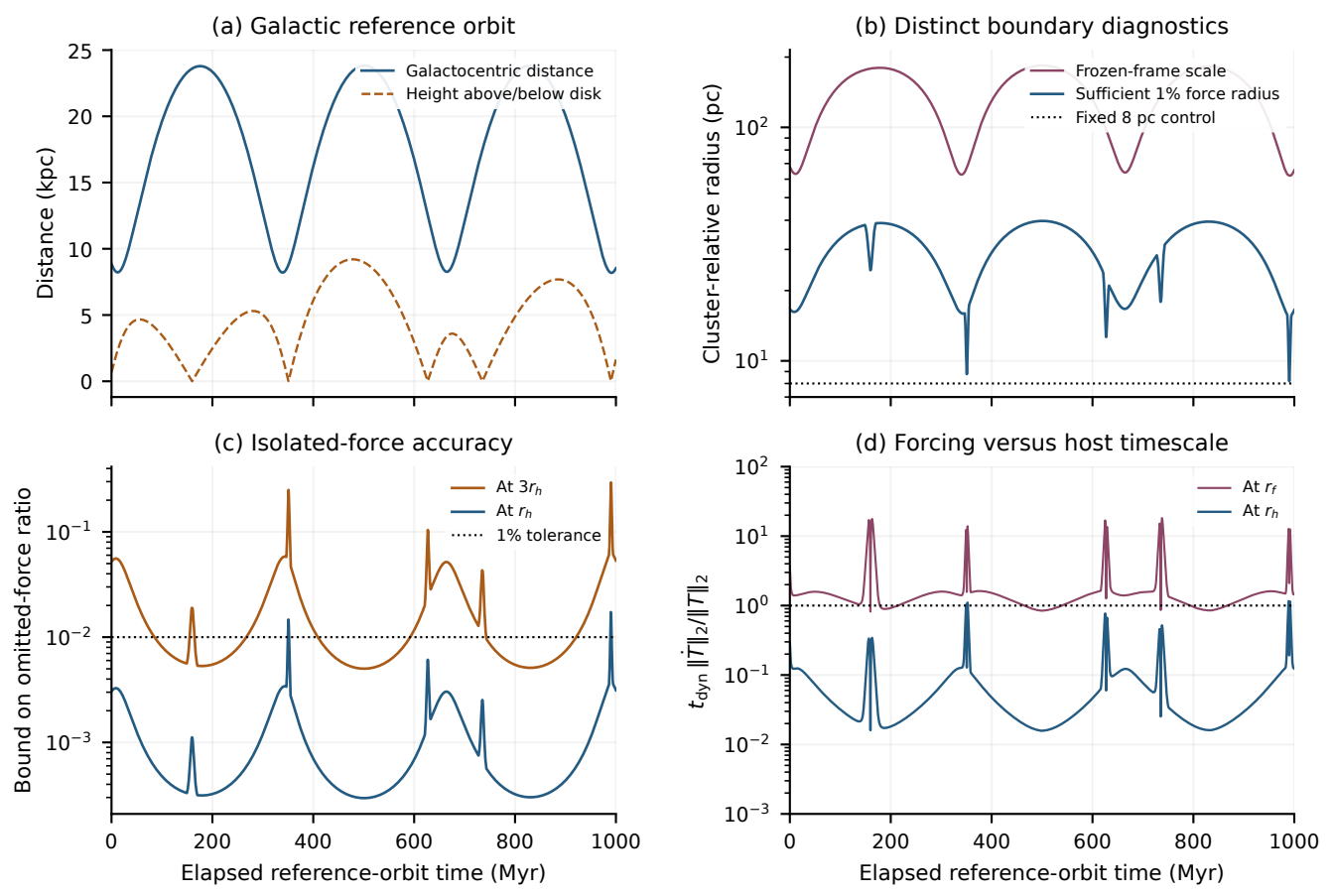
(a) Galactic reference orbit
Distance (kpc)
Galactocentric distance
Height above/below disk
(b) Distinct boundary diagnostics
Cluster-relative radius (pc)
Frozen-frame scale
Sufficient 1% force radius
Fixed 8 pc control
(c) Isolated-force accuracy
Bound on omitted-force ratio
At $3r_h$
At r_h
1% tolerance
Elapsed reference-orbit time (Myr)
(d) Forcing versus host timescale
$t_{\mathrm{dyn}} \|\dot{T}\|_2/\|T\|_2$
At r_f
At r_h
Elapsed reference-orbit time (Myr)